Rational and Integral Points on $y^2 - x^2y + z^2 + 1 = 0$

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Abstract

We give a complete, unconditional, and self-contained description of the rational solutions of

$$y^2 - x^2y + z^2 + 1 = 0.$$

If a nonzero rational number is written in lowest terms as $x = \varepsilon a/b$, where $a, b > 0$, $\gcd(a, b) = 1$, and $\varepsilon \in \{\pm 1\}$, then the fibre above x has a rational point if and only if $N_{a,b} = a^4 - 4b^4$ is positive and every prime congruent to 3 modulo 4 occurs in $N_{a,b}$ with even exponent. Whenever this criterion holds, an integral representation $N_{a,b} = C^2 + D^2$ exists, and a homogeneous rational parametrisation of the entire fibre is written down explicitly. We then prove that every admissible reduced fraction a/b has odd numerator and even denominator. It follows, more strongly than the absence of integral triples, that no rational solution can have integral x. A second independent congruence proof of the integral obstruction is included. Finally, we exhibit both a complete one-parameter fibre over $x = \pm 3/2$ and a Pell recurrence producing infinitely many distinct rational x-coordinates.

1 Statement of the classification

Let

$$\mathcal{S}(F) = \{(x, y, z) \in F^3 : y^2 - x^2y + z^2 + 1 = 0\}$$

for a subfield $F \subseteq \mathbb{R}$. For coprime positive integers a, b, put

$$N_{a,b} = a^4 - 4b^4.$$

For a prime p and a nonzero integer n, let $\mathrm{ord}_p(n)$ denote the exponent of p in the prime factorisation of $|n|$.

Theorem 1.1 (Complete rational classification)**.** *Let $a, b \in \mathbb{Z}_{>0}$ satisfy $\gcd(a, b) = 1$, and let $\varepsilon \in \{\pm 1\}$. The fibre*

$$\mathcal{S}_{\varepsilon a/b}(\mathbb{Q}) = \{(y, z) \in \mathbb{Q}^2 : (\varepsilon a/b, y, z) \in \mathcal{S}(\mathbb{Q})\}$$

is nonempty if and only if

$$N_{a,b} > 0 \quad \textit{and} \quad \mathrm{ord}_q(N_{a,b}) \equiv 0 \pmod 2 \quad \textit{for every prime } q \equiv 3 \pmod 4. \tag{1.1}$$

Assume (1.1)*. Choose integers C, D such that*

$$C^2 + D^2 = N_{a,b}. \tag{1.2}$$

For $[u:v] \in \mathbb{P}^1(\mathbb{Q})$*, set*

$$U = u^2 + v^2.$$

Then $U \neq 0$*, and every rational point in the fibre, exactly once, is obtained from*

$$x = \varepsilon \frac{a}{b}, \tag{1.3}$$

$$y = \frac{a^2 U + C(u^2 - v^2) - 2Duv}{2b^2 U}, \tag{1.4}$$

$$z = \frac{D(u^2 - v^2) + 2Cuv}{2b^2 U}. \tag{1.5}$$

Here "exactly once" refers to a fixed choice of a, b, ε, C, D *and the usual projective identification* $[u : v] = [\lambda u : \lambda v]$ *for* $\lambda \in \mathbb{Q}^\times$*.*

Moreover, every pair (a, b) *satisfying* (1.1) *obeys*

$$a \equiv 1 \pmod 2, \qquad b \equiv 0 \pmod 2. \tag{1.6}$$

Consequently no point of $\mathcal{S}(\mathbb{Q})$ *has integral* x*, and in particular*

$$\mathcal{S}(\mathbb{Z}) = \varnothing.$$

The proof is assembled from elementary identities, the two-square criterion, and the rational parametrisation of a circle. All of these ingredients are proved below.

2 Elementary reductions

Proposition 2.1 (Completion of the square)**.** *For all real* x, y, z *one has the identity*

$$4(y^2 - x^2 y + z^2 + 1) = (2y - x^2)^2 + (2z)^2 - (x^4 - 4). \tag{2.1}$$

Thus

$$y^2 - x^2 y + z^2 + 1 = 0 \iff (2y - x^2)^2 + (2z)^2 = x^4 - 4. \tag{2.2}$$

Every rational solution satisfies

$$0 < y < x^2 \qquad \text{and} \qquad x^4 > 4. \tag{2.3}$$

Proof. Expanding the right-hand side of (2.1) gives

$$4y^2 - 4x^2 y + x^4 + 4z^2 - x^4 + 4,$$

which is the left-hand side. This proves (2.2).

If the original equation holds, then

$$y(x^2 - y) = z^2 + 1 > 0. \tag{2.4}$$

If $y \leq 0$, then $x^2 - y \geq 0$, so the left-hand side of (2.4) is nonpositive, a contradiction. Hence $y > 0$; then (2.4) forces $x^2 - y > 0$. This proves $0 < y < x^2$.

Equation (2.2) gives $x^4 \geq 4$. Equality would force $2y - x^2 = 0$ and $z = 0$, and hence $x^2 = 2$. A rational number cannot have square 2: if $x = r/s$ in lowest terms and $r^2 = 2s^2$, then r is even, after which s is also even, contradicting $\gcd(r, s) = 1$. Therefore a rational solution has $x^4 > 4$. □

3 The two-square criterion over $\mathbb{Q}$

We prove the precise arithmetic criterion needed later. The argument also proves that allowing rational squares does not enlarge the class of positive integers representable as a sum of two squares.

Lemma 3.1. *Let $q \equiv 3 \pmod 4$ be prime. If $A, B \in \mathbb{Z}$ and $q \mid A^2 + B^2$, then $q \mid A$ and $q \mid B$.*

Proof. If $q \mid B$, then $q \mid A^2$ and hence $q \mid A$. Suppose instead that $q \nmid B$. Modulo q, let $r \equiv AB^{-1}$. Then $r^2 \equiv -1 \pmod q$.

For completeness, multiplication by the nonzero residue r permutes the nonzero residues modulo q. Multiplying all these residues and cancelling the nonzero product gives $r^{q-1} \equiv 1 \pmod q$. On the other hand,

$$r^{q-1} = (r^2)^{(q-1)/2} \equiv (-1)^{(q-1)/2} \equiv -1 \pmod q,$$

because $(q-1)/2$ is odd. This contradiction proves that $q \mid B$, and then also $q \mid A$. □

Corollary 3.2. *If $q \equiv 3 \pmod 4$ is prime and $A, B \in \mathbb{Z}$ are not both zero, then*

$$\operatorname{ord}_q(A^2 + B^2)$$

is even.

Proof. Let

$$e = \min\{\operatorname{ord}_q(A), \operatorname{ord}_q(B)\},$$

where $\operatorname{ord}_q(0)$ may be regarded as $+\infty$. Write $A = q^e A_0$ and $B = q^e B_0$, with at least one of A_0, B_0 not divisible by q. By Lemma 3.1, $q \nmid A_0^2 + B_0^2$. Therefore

$$\operatorname{ord}_q(A^2 + B^2) = 2e.$$

□

Lemma 3.3 (Primes congruent to 1 modulo 4)**.** *Every prime $p \equiv 1 \pmod 4$ can be written as a sum of two integer squares.*

Proof. Write $p = 4k + 1$ and put

$$m = \frac{p-1}{2} = 2k.$$

We first produce a square root of -1 modulo p. Wilson's congruence

$$(p-1)! \equiv -1 \pmod p \tag{3.1}$$

follows by pairing every nonzero residue with its multiplicative inverse: the only self-inverse nonzero residues are 1 and -1, since $t^2 \equiv 1 \pmod p$ implies $(t-1)(t+1) \equiv 0 \pmod p$.

Pairing j with $p - j$ for $1 \le j \le m$, and using that m is even, we obtain

$$(p-1)! = \prod_{j=1}^{m} j(p-j) \equiv (-1)^m (m!)^2 = (m!)^2 \pmod p.$$

Together with (3.1), this gives

$$r^2 \equiv -1 \pmod p, \qquad r \equiv m! \pmod p. \tag{3.2}$$

Let $M = \lfloor \sqrt{p} \rfloor$. Since a prime is not a perfect square,

$$M^2 < p < (M+1)^2.$$

There are $(M+1)^2 > p$ pairs (u, v) with $0 \le u, v \le M$, but only p residue classes of $u - rv$ modulo p. By the pigeonhole principle, two distinct such pairs have the same residue. Subtracting them yields integers A, B, not both zero, such that

$$|A| \le M, \qquad |B| \le M, \qquad A \equiv rB \pmod{p}.$$

By (3.2),

$$A^2 + B^2 \equiv (r^2+1)B^2 \equiv 0 \pmod{p}.$$

Moreover,

$$0 < A^2 + B^2 \le 2M^2 < 2p.$$

Thus $A^2 + B^2$ is a positive multiple of p strictly smaller than $2p$, so it equals p. ☐

Theorem 3.4 (Two-square criterion)**.** *For a positive integer N, the following conditions are equivalent:*

(i) *N is a sum of two rational squares;*

(ii) *N is a sum of two integer squares;*

(iii) *$\operatorname{ord}_q(N)$ is even for every prime $q \equiv 3 \pmod 4$.*

Proof. Condition (ii) immediately implies (i). Suppose (i) holds, say

$$N = \left(\frac{A}{M}\right)^2 + \left(\frac{B}{M}\right)^2$$

for integers A, B and a positive integer M. Then

$$NM^2 = A^2 + B^2.$$

For every prime $q \equiv 3 \pmod 4$, Corollary 3.2 gives

$$\operatorname{ord}_q(N) + 2\operatorname{ord}_q(M) = \operatorname{ord}_q(A^2+B^2) \equiv 0 \pmod{2}.$$

Hence $\operatorname{ord}_q(N)$ is even, proving (iii).

Now assume (iii). In the prime factorisation of N, every prime $q \equiv 3 \pmod 4$ therefore occurs as q^{2e}, which is $(q^e)^2 + 0^2$. Also $2 = 1^2 + 1^2$, and every prime $p \equiv 1 \pmod 4$ is a sum of two integer squares by Lemma 3.3. Finally, the identity

$$(A^2+B^2)(C^2+D^2) = (AC - BD)^2 + (AD + BC)^2 \tag{3.3}$$

shows that a product of sums of two integer squares is again such a sum. Repeated application of (3.3) to the prime factorisation of N proves (ii). ☐

4 A rational parametrisation of every circle fibre

Lemma 4.1 (The rational unit circle)**.** *The map*

$$\Phi : \mathbb{P}^1(\mathbb{Q}) \longrightarrow \{(A,B) \in \mathbb{Q}^2 : A^2 + B^2 = 1\}, \qquad [u:v] \longmapsto \left(\frac{u^2 - v^2}{u^2 + v^2}, \frac{2uv}{u^2 + v^2} \right) \tag{4.1}$$

is a bijection.

Proof. A nonzero rational pair (u,v) cannot satisfy $u^2 + v^2 = 0$, so the map is well-defined. It is invariant under replacing (u,v) by $(\lambda u, \lambda v)$ with $\lambda \neq 0$, and direct expansion shows that its image lies on the unit circle.

Conversely, let $(A,B) \in \mathbb{Q}^2$ satisfy $A^2 + B^2 = 1$. If $A = -1$, then $B = 0$, and $(A,B) = \Phi([0:1])$. If $A \neq -1$, take

$$[u:v] = [1 + A : B].$$

Since

$$(1 + A)^2 + B^2 = 2(1 + A),$$

a direct calculation gives

$$\frac{(1 + A)^2 - B^2}{(1 + A)^2 + B^2} = A, \qquad \frac{2(1 + A)B}{(1 + A)^2 + B^2} = B.$$

This also gives an explicit inverse to Φ, so the map is bijective. $\square$

Lemma 4.2 (A circle with a rational base point)**.** *Let $N > 0$, and let $C, D \in \mathbb{Q}$ satisfy $C^2 + D^2 = N$. Then the map*

$$\Psi_{C,D} : \mathbb{P}^1(\mathbb{Q}) \longrightarrow \{(R,S) \in \mathbb{Q}^2 : R^2 + S^2 = N\}, \tag{4.2}$$

$$[u:v] \longmapsto \left(\frac{C(u^2 - v^2) - 2Duv}{u^2 + v^2}, \frac{D(u^2 - v^2) + 2Cuv}{u^2 + v^2} \right) \tag{4.3}$$

is a bijection.

Proof. Let $(A,B) = \Phi([u:v])$ as in Lemma 4.1. The displayed map is equivalently

$$\begin{pmatrix} R \\ S \end{pmatrix} = \begin{pmatrix} C & -D \\ D & C \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix}. \tag{4.4}$$

Consequently

$$R^2 + S^2 = (C^2 + D^2)(A^2 + B^2) = N.$$

Conversely, if $R^2 + S^2 = N$, define

$$A = \frac{CR + DS}{N}, \qquad B = \frac{CS - DR}{N}. \tag{4.5}$$

Then $A, B \in \mathbb{Q}$ and

$$A^2 + B^2 = \frac{(C^2 + D^2)(R^2 + S^2)}{N^2} = 1.$$

Equations (4.4) and (4.5) are inverse transformations. The result therefore follows from the bijectivity of Φ. $\square$

5 Proof of the complete classification

Proposition 5.1 (Bijection with a two-square equation)**.** *Let $a, b \in \mathbb{Z}_{>0}$ be coprime, let $\varepsilon \in \{\pm 1\}$, and put $x = \varepsilon a/b$. The assignments*

$$R = b^2(2y - x^2) = 2b^2y - a^2, \qquad S = 2b^2z \tag{5.1}$$

give a bijection

$$\mathcal{S}_{\varepsilon a/b}(\mathbb{Q}) \longleftrightarrow \{(R,S) \in \mathbb{Q}^2 : R^2 + S^2 = N_{a,b}\}.$$

The inverse map is

$$y = \frac{a^2 + R}{2b^2}, \qquad z = \frac{S}{2b^2}. \tag{5.2}$$

Proof. Multiplying (2.2) by b^4 gives

$$\left(b^2(2y - x^2)\right)^2 + (2b^2z)^2 = b^4(x^4 - 4) = a^4 - 4b^4.$$

This is precisely $R^2 + S^2 = N_{a,b}$. The formulas in (5.1) are visibly inverted by (5.2), proving the bijection. □

Proof of Theorem 1.1. By Proposition 5.1, the fibre is nonempty if and only if the positive integer $N_{a,b}$ is a sum of two rational squares. The positivity is necessary because Proposition 2.1 shows $x^4 > 4$. Theorem 3.4 proves that nonemptiness is equivalent to (1.1), and also guarantees integers C, D satisfying (1.2).

For such C, D, Lemma 4.2 parametrises every pair (R, S) with $R^2 + S^2 = N_{a,b}$ exactly once. Substitution into (5.2) gives precisely (1.4)–(1.5). Thus the asserted formulas are complete and introduce no extraneous solutions.

It remains to prove the parity assertion (1.6); this is Proposition 6.1 below. Once it is known, a rational solution with integral x is impossible, because the reduced denominator of an integer is 1, whereas every admissible denominator is even. In particular no triple in $\mathcal{S}(\mathbb{Z})$ exists. □

6 The numerator–denominator obstruction

We use the elementary observation that every positive integer congruent to 3 modulo 4 has at least one prime congruent to 3 modulo 4 occurring to odd exponent. Indeed, if all such exponents were even, its odd part would be congruent to 1 modulo 4.

Proposition 6.1. *Let $a, b \in \mathbb{Z}_{>0}$ be coprime. If $N_{a,b} > 0$ and every prime $q \equiv 3 \pmod 4$ occurs in $N_{a,b}$ with even exponent, then a is odd and b is even.*

Proof. Suppose first that a is even. Coprimality forces b to be odd. Write $a = 2k$. Then

$$N_{a,b} = 4(4k^4 - b^4). \tag{6.1}$$

The positive odd integer $4k^4 - b^4$ is congruent to 3 modulo 4. It therefore contains a prime $q \equiv 3 \pmod 4$ to odd exponent. Since q is odd, (6.1) shows that the same exponent occurs in $N_{a,b}$, a contradiction.

It remains to exclude the possibility that both a and b are odd. Since $N_{a,b} > 0$, one has $a^2 > 2b^2$, and

$$N_{a,b} = (a^2 - 2b^2)(a^2 + 2b^2). \tag{6.2}$$

Both factors are positive and congruent to 3 modulo 4. They are coprime. Indeed, any common divisor d is odd and divides

$$(a^2+2b^2)+(a^2-2b^2)=2a^2$$

and

$$(a^2+2b^2)-(a^2-2b^2)=4b^2.$$

Thus $d \mid a^2$ and $d \mid b^2$, whence $d=1$ because $\gcd(a,b)=1$.

Each factor in (6.2), being 3 modulo 4, contains a prime congruent to 3 modulo 4 to odd exponent. Coprimality of the factors prevents that exponent from being altered by the other factor. Hence $N_{a,b}$ has a prime congruent to 3 modulo 4 to odd exponent, again a contradiction. Therefore a is odd and b is even. □

Corollary 6.2 (Factorwise form of the criterion)**.** *Let $a,b \in \mathbb{Z}_{>0}$ be coprime. The fibre above $x=\pm a/b$ is nonempty over $\mathbb{Q}$ if and only if*

(a) *a is odd and b is even;*

(b) *$a^2>2b^2$;*

(c) *each of a^2-2b^2 and a^2+2b^2 is a sum of two integer squares.*

Proof. Assume first that the fibre is nonempty. Conditions (a) and (b) follow from Proposition 6.1 and $N_{a,b}>0$. Put

$$A=a^2-2b^2, \qquad B=a^2+2b^2.$$

Because a is odd and b is even, both A and B are odd. The same greatest-common-divisor argument used in Proposition 6.1 shows $\gcd(A,B)=1$. Since $AB=N_{a,b}$ and every prime $q\equiv 3 \pmod 4$ occurs in AB with even exponent, coprimality implies that it occurs with even exponent in each of A and B. Theorem 3.4 proves (c).

Conversely, if (a)–(c) hold, then $N_{a,b}=AB>0$, and the product identity (3.3) shows that $N_{a,b}$ is a sum of two integer squares. Proposition 5.1 then gives a rational point in the fibre. □

Corollary 6.3. *No rational solution has $x\in\mathbb{Z}$. Consequently*

$$\mathcal{S}(\mathbb{Z})=\varnothing.$$

Proof. If $x\in\mathbb{Z}$, its expression $x=\varepsilon a/b$ in lowest terms has $b=1$. Proposition 6.1 requires b to be even, which is impossible. □

7 An independent congruence proof of the integral obstruction

The preceding proof gives the stronger statement that integral x is impossible even when y and z are merely rational. We record a second proof of the weaker assertion $\mathcal{S}(\mathbb{Z})=\varnothing$ that uses only divisibility and congruences.

Lemma 7.1. *Let $z\in\mathbb{Z}$, and let $d>0$ divide z^2+1. Then*

$$d=2^{\delta}m, \qquad \delta\in\{0,1\}, \tag{7.1}$$

where every prime divisor of the odd integer m is congruent to 1 modulo 4. In particular, $m\equiv 1 \pmod 4$.

Proof. Let p be an odd prime divisor of z^2+1. Then $p \nmid z$ and $z^2 \equiv -1 \pmod{p}$. If $p \equiv 3 \pmod{4}$, the argument in Lemma 3.1, applied to z^2+1^2, would force $p \mid 1$, which is impossible. Hence $p \equiv 1 \pmod{4}$.

For the power of 2, if z is even then z^2+1 is odd. If z is odd, then $z^2 \equiv 1 \pmod{8}$, so $z^2+1 \equiv 2 \pmod{8}$ and its 2-adic valuation is exactly 1. Every positive divisor therefore has the form (7.1). Since all prime divisors of m are 1 modulo 4, so is m. ☐

Theorem 7.2 (Direct integral obstruction)**.** *There are no integers x, y, z satisfying*

$$y^2 - x^2y + z^2 + 1 = 0.$$

Proof. Assume an integral solution exists. Equation (2.4) shows that $y > 0$ and

$$y \mid z^2 + 1.$$

By Lemma 7.1, write $y = 2^\delta m$, where $\delta \in \{0, 1\}$ and $m \equiv 1 \pmod{4}$.

First suppose $\delta = 0$. Then y is odd and $y \equiv 1 \pmod{4}$. Rearrange the equation as

$$x^2y = y^2 + z^2 + 1. \tag{7.2}$$

If x were even, the left side would be 0 modulo 4, whereas the right side would be 2 or 3 modulo 4, according as z is even or odd. Thus x is odd. If z were even, the right side of (7.2) would be 2 modulo 4, while the left side would be odd. Hence z is odd. Reducing (7.2) modulo 4 now gives

$$y \equiv 1 + 1 + 1 \equiv 3 \pmod{4},$$

contradicting $y \equiv 1 \pmod{4}$.

Now suppose $\delta = 1$. Then $y = 2m \equiv 2 \pmod{8}$. Since y divides z^2+1, the latter is even, so z is odd. Reducing the right side of (7.2) modulo 8 gives

$$y^2 + z^2 + 1 \equiv 4 + 1 + 1 \equiv 6 \pmod{8}.$$

If x is even, then $x^2y \equiv 0 \pmod{8}$; if x is odd, then $x^2y \equiv 2 \pmod{8}$. Neither possibility is 6 modulo 8, a final contradiction. ☐

8 Explicit infinite families

Example 8.1 (The complete fibre over $x = \pm 3/2$)**.** Take $a = 3$ and $b = 2$. Then

$$N_{3,2} = 3^4 - 4 \cdot 2^4 = 17 = 1^2 + 4^2.$$

With $C = 1$, $D = 4$, and $[u : v] = [1 : t]$, formulas (1.4)–(1.5) become

$$x = \pm\frac{3}{2}, \tag{8.1}$$

$$y = \frac{4t^2 - 4t + 5}{4(t^2+1)}, \tag{8.2}$$

$$z = \frac{-2t^2 + t + 2}{4(t^2+1)}, \tag{8.3}$$

where $t \in \mathbb{Q}$. The missing projective parameter $[0 : 1]$, customarily written $t = \infty$, gives

$$(x, y, z) = \left(\pm\frac{3}{2}, 1, -\frac{1}{2}\right).$$

By Theorem 1.1, these formulas give not merely infinitely many examples but every rational point in either fibre $x = 3/2$ or $x = -3/2$. For instance, $t = 0$ gives

$$\left(\frac{3}{2}, \frac{5}{4}, \frac{1}{2}\right).$$

The preceding example already proves that $\mathcal{S}(\mathbb{Q})$ is infinite. The next construction shows that infinitely many distinct rational x-coordinates occur.

Proposition 8.2 (A Pell subfamily). *Define positive integer pairs (a_n, b_n) recursively by*

$$(a_1, b_1) = (3, 2), \qquad a_{n+1} = 3a_n + 4b_n, \qquad b_{n+1} = 2a_n + 3b_n. \tag{8.4}$$

Then, for every $n \geq 1$,

$$a_n^2 - 2b_n^2 = 1, \tag{8.5}$$

$\gcd(a_n, b_n) = 1$*, and the fibre above $x = \pm a_n/b_n$ is nonempty. More explicitly, for every $[u : v] \in \mathbb{P}^1(\mathbb{Q})$, with $U = u^2 + v^2$,*

$$x = \pm\frac{a_n}{b_n}, \tag{8.6}$$

$$y = \frac{a_n^2 U + (u^2 - v^2) - 4b_n uv}{2b_n^2 U}, \tag{8.7}$$

$$z = \frac{2b_n(u^2 - v^2) + 2uv}{2b_n^2 U} \tag{8.8}$$

solves the equation. The positive numbers a_n/b_n are pairwise distinct.

Proof. A direct calculation shows that the recurrence preserves the quadratic form $a^2 - 2b^2$:

$$\begin{aligned}
&(3a + 4b)^2 - 2(2a + 3b)^2 \\
&\quad = 9a^2 + 24ab + 16b^2 - 8a^2 - 24ab - 18b^2 \\
&\quad = a^2 - 2b^2.
\end{aligned}$$

Since $3^2 - 2 \cdot 2^2 = 1$, induction proves (8.5). Any common divisor of a_n and b_n would have square dividing the left side of (8.5), so it must be 1. The recurrence also gives $b_{n+1} = 2a_n + 3b_n > b_n$, so the sequence (b_n) is strictly increasing.

From (8.5),

$$\begin{aligned}
N_{a_n,b_n} &= (a_n^2 - 2b_n^2)(a_n^2 + 2b_n^2) \\
&= 1 \cdot (1 + 4b_n^2) = 1^2 + (2b_n)^2.
\end{aligned} \tag{8.9}$$

Thus one may choose $C = 1$ and $D = 2b_n$ in Theorem 1.1; formulas (8.6)–(8.8) are exactly the resulting parametrisation.

Finally,

$$\left(\frac{a_n}{b_n}\right)^2 = 2 + \frac{1}{b_n^2}.$$

Since the positive integers b_n strictly increase, these squares, and hence the positive ratios a_n/b_n, are pairwise distinct. □

The first few pairs are

$$(a_n, b_n) = (3, 2),\ (17, 12),\ (99, 70),\ (577, 408), \ldots.$$

Thus rational points occur on infinitely many distinct circle fibres, while every such x necessarily has an even reduced denominator.